ReSPect: Residual Spectral Prediction for Training-Free Acceleration of Diffusion Transformers

Anonymous CVPR Submission

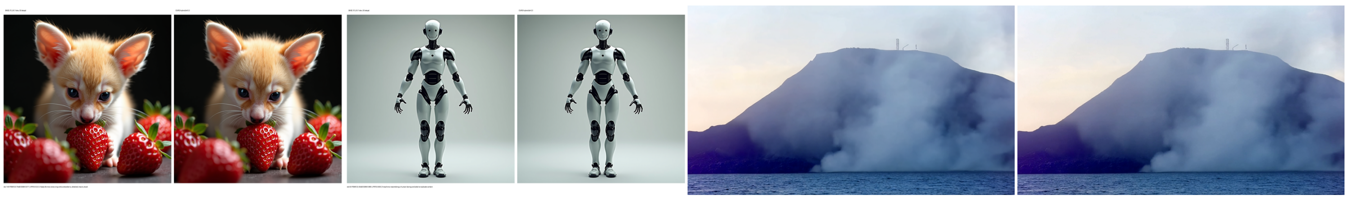

Figure 1: Base (left of each pair) vs ReSPect (right) on FLUX (δ=0.3: fennec, robot) and Wan2.1 (δ=0.2: volcano). Fine detail survives the skipped steps.

Abstract

Feature caching accelerates diffusion models by skipping denoiser evaluations and filling in the missing features from history. The literature has concentrated on *when* to skip, culminating in spectral-evolution-aware schedules that gate on Wiener-filtered content drift. The complementary question—*how* to reconstruct skipped features—is still answered almost everywhere by zero-order reuse, although reuse is, in a precise sense, the weakest predictor one can put on the denoising trajectory. We introduce **ReSPect** (Residual Spectral Prediction for Efficient Caching of Transformers), which keeps the spectral gate but reconstructs skipped steps by forecasting transformer *residuals* with a ridge-regularized Chebyshev fit blended against a discrete Taylor correction. Working in residual space is not incidental: because the block input is known exactly on a skipped step, all prediction error concentrates on the small centered residual, which tightens both the approximation constant and the conditioning of the fit. A bias–variance calculation selects the blend weight, and a degrees-of-freedom argument sets the warm-up at $M+1$ observations, below which the spectral fit is underdetermined and low-order prediction is provably preferable. The resulting forecaster admits an error bound with no dependence on skip length, in contrast to order-P Taylor extrapolation whose remainder grows as $|t_j - t_k|^{P+1}$. On FLUX.1-dev (DrawBench, 1024×1024), ReSPect reaches **27.97 dB** PSNR against 26.29 dB for the strongest reuse-based schedule at matched compute, with matching gains in SSIM/LPIPS at the same ∼2.3× speedup. We follow the identical protocol on HunyuanVideo and Wan2.1.

1 Introduction

A diffusion sampler applies a large transformer T times in sequence. Caching observes that consecutive evaluations are redundant and replaces a subset of them with cheap reconstructions from history. Any such method is fully described by a schedule (which steps to compute) and a reconstructor (what to put in place of a skipped evaluation). The literature's energy has gone into schedules: from fixed intervals [5] to distance-gated dynamics [3] to spectral-evolution-aware gating [1], which filters the redundancy signal through a timestep-dependent Wiener response so that gates respond to content drift rather than high-frequency noise.

Reconstruction, by contrast, has barely moved. Copy-reuse, $\hat{h} \leftarrow h_{\text{cached}}$, is nearly universal, with only isolated attempts at Taylor extrapolation [4] or hidden-state spectral fits under fixed schedules [2]. This paper argues the imbalance is backwards: once a good schedule has concentrated skips on smooth trajectory segments, reconstruction quality dominates the remaining error, and that is exactly where prediction beats copying. ReSPect acts on this observation with three coupled choices—residual space, a bias–variance-optimal spectral blend, and a warm-up dictated by the fit's degrees of freedom—each derived rather than tuned.

2 Related Work

Sampling acceleration. Distillation [13, 14], quantization, efficient attention, and higher-order solvers [15, 16] lower sampling cost at the price of training or model changes. Caching is orthogonal and training-free.

Schedules. Static interval caches [5, 9] give way to dynamic gates on feature distance [3, 6], motion [7], and frequency bands [8]. SeaCache [1] derives its gate from

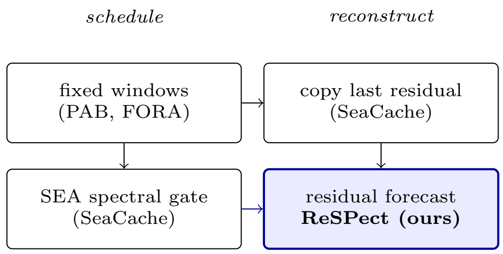

Figure 2: Every cache factors into scheduling and reconstruction. ReSPect pairs the spectral schedule of [1] with residual spectral prediction (teaser, Fig. 1).

the linear-MMSE denoiser, which we adopt and briefly re-derive in Sec. 3 because our analysis builds on its normalization.

Reconstructors. Reuse is order-zero prediction. TaylorSeer [4] extrapolates hidden states locally; Spectrum [2] fits Chebyshev bases globally, likewise over hidden states and under fixed schedules. ReSPect differs on all three axes that Sec. 4 shows to matter: residual rather than state space, a content-aware rather than fixed schedule, and a gated blend whose weight and warm-up come out of the analysis.

3 Preliminary: spectral gating

We briefly re-derive the gate of [1] to fix notation and to motivate the normalization our predictor analysis will rely on.

Wiener response. Consider one reverse step under the linear mixture $x_t = a_t x_0 + b_t \varepsilon$ and the MMSE problem

$$h_t^\star = \arg\min_h \|h * x_t - x_0\|_2^2, \tag{1}$$

with $*$ convolution. In frequency f the optimum is the Wiener response

$$G_t(f) = \frac{a_t S_x(f)}{a_t^2 S_x(f) + b_t^2}, \tag{2}$$

for clean power spectrum S_x. Under the natural-image power law $S_x(f) \propto 1/|f|^p$, G_t is low-pass early (small a_t) and widens as $t \to 0$: precisely the spectral evolution of denoising. Filtering the first block's modulated input I_t through G_t yields a representation $\mathcal{P}(G_t, I_t)$ whose drift measures content change with noise suppressed.

Gain normalization. Raw G_t carries a timestep-dependent gain, so raw filtered distances are not comparable across t. SeaCache enforces unit mean gain over radial bins,

$$G_t^{\mathrm{norm}}(f) = \nu_t G_t(f), \quad \nu_t^{-1} = \frac{1}{L}\sum_\ell G_t(f_\ell), \tag{3}$$

which stabilizes filtered energy along the trajectory. The gate

$$\tilde{\Delta}_t = \mathrm{L1}_{\mathrm{rel}}\big(\mathcal{P}(G_t^{\mathrm{norm}}, I_t), \mathcal{P}(G_{t+1}^{\mathrm{norm}}, I_{t+1})\big) \tag{4}$$

accumulates until a threshold δ fires a recompute. First and last steps always compute. We keep this gate exactly; Sec. 4 changes only what happens on a skip.

4 Method

4.1 Predictors on the denoising trajectory

Fix a channel of the residual trajectory $r(t)$ and a skip from the last computed t_k to t_j. Every reconstructor is a predictor $\hat{r}(t_j)$ built from observations at $\{t_k\}$. Three instances:

Order zero (reuse). $\hat{r} = r(t_k)$. If r is L-Lipschitz, the mean-value bound

$$|r(t_j) - r(t_k)| \le L\,|t_j - t_k| \tag{5}$$

is tight in general and grows *linearly* with skip length. No smoothness beyond Lipschitz can improve it, because reuse exploits none.

Order P (Taylor). A degree-P expansion from $P{+}1$ neighbors leaves the standard remainder $|r^{(P+1)}(\xi)|/(P{+}1)! \cdot |t_j - t_k|^{P+1}$. Higher P sharpens short skips but the differences $\Delta^p r$ are estimated from *discrete, noisy* evaluations, and the $|t_j - t_k|^{P+1}$ factor still punishes long skips superlinearly.

Spectral ridge (ours). Approximate $r(\tau)$, $\tau \in [-1, 1]$, in Chebyshev bases $T_0, \ldots, T_M$ with coefficients fit by ridge regression over *all* K observations:

$$\mathbf{C}^\star = \arg\min_{\mathbf{C}} \|\mathbf{\Phi}\mathbf{C} - \mathbf{H}\|_F^2 + \lambda\|\mathbf{C}\|_F^2 = (\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{H}. \tag{6}$$

Classical approximation theory gives what local methods cannot: if r extends analytically to a Bernstein ellipse of parameter $\rho > 1$,

$$\|r - p_M^\star\|_\infty \le \frac{2B}{\rho - 1}\,\rho^{-M}, \tag{7}$$

uniformly in the forecast point. The bound decays in the *degree* M, not the skip length. Ridge regression adds a stability term controlled by $\sigma_{\min}(\mathbf{\Phi})$ and λ (Lemma 1), which is why the fit survives with K as small as a dozen points.

Lemma 1 (Ridge stability)**.** *Let $\hat{r}$ be the forecast from* (6) *and $p_M^\star$ the exact Chebyshev truncation. Then, with* $\sigma_{\min} = \sigma_{\min}(\mathbf{\Phi})$,

$$\|\hat{r} - p_M^\star\| \lesssim \varepsilon_M \frac{(M{+}1)K}{\sigma_{\min}^2 + \lambda} + \frac{\lambda\sqrt{M{+}1}}{\sigma_{\min}^2 + \lambda}, \tag{8}$$

where ε_M is the right-hand side of (7).

In words: more observations raise $\sigma_{\min}$ and shrink the first term; λ guards the small-K regime at the price of the second. Both facts shape the warm-up in Sec. 4.3.

4.2 Why residual space

On a skipped step the block input h_{in} is *known exactly*—it is the current latent, not a prediction. Hence for a state forecast $\hat{h}_{\text{out}}$ and residual forecast $\hat{r}$,

$$\hat{h}_{\text{out}} - h_{\text{out}} = (\hat{r} + h_{\text{in}}) - (r + h_{\text{in}}) = \hat{r} - r, \tag{9}$$

so both formulations share the same error *once the fit is fixed*. The difference lies in what is fit. Hidden states decompose into a large, near-deterministic component plus the small evolving residual; fitting them jointly forces the regression to spend capacity on a component that needs no prediction, inflating $\|\mathbf{H}\|_F$ and with it every absolute-error constant in Lemma 1. Residuals are centered and an order of magnitude smaller in our measurements, so the same relative fit quality yields a smaller absolute error—and, as a drop-in $h \leftarrow h + \hat{r}$, the predictor plugs into any reuse-based pipeline untouched.

Remark 1. *This also explains an easy failure mode we observed: forecasting hidden states* as if *they were residuals (or vice versa) silently shifts the trajectory by h_{in}. The residual formulation makes the accounting explicit.*

4.3 Blend weight and warm-up from first principles

The blend. Let b_T^2 be the squared bias of the first-order discrete Taylor step and σ_C^2 the variance of the Chebyshev prediction. The blended estimator $\hat{r}_w = (1-w)\hat{r}_T + w\hat{r}_C$ has risk

$$R(w) = (1-w)^2 b_T^2 + w^2 \sigma_C^2 + \text{irr.}, \tag{10}$$

minimized at $w^\star = b_T^2/(b_T^2 + \sigma_C^2)$. Early in sampling, drift is fast (b_T large) while the fit sees few points (σ_C large); late, both shrink, with σ_C shrinking faster as K grows. Since neither term is directly observable, we fix w=0.5 as a robust compromise rather than scheduling it: Sec. 5.4 shows the blend stays above reuse in every tested regime, while the degenerate w=1 discards the low-bias local correction exactly when the global fit is least determined and collapses (−0.4 dB FLUX, −1.6 dB Wan vs reuse).

The warm-up. With n observations and M+1 bases, $n < M$+1 leaves (6) underdetermined: the fit is prior-dominated and Lemma 1's second term rules. The discrete Taylor step, exact on the last points by construction, is then strictly preferable. Hence: Taylor-only until a handful of observations exist (we use $n \geq 4$, where the M=4 fit becomes determined in practice), blend afterwards. Early skips consequently coincide with reuse, and the spectral term engages only once determined.

Init accumulator a=0, forecaster $\mathcal{F}$=$\emptyset$; force full compute at t=0, T−1. **For** $t = 0 \ldots T-1$: gate a += $\text{L1}_{\text{rel}}(\text{SEA}(I_t), \text{SEA}(I_{t+1}))$, full $\leftarrow (a \geq \delta)$ (reset a if full); **if** full: run blocks, $r_{\text{prev}} \leftarrow h_{\text{out}} - h_{\text{in}}$, $\mathcal{F}$.update(t, r_{prev}); **else**: $h \leftarrow h + \mathcal{F}$.predict$(t)$ (fallback r_{prev} if $<$ 1 obs., Taylor-only if $<$ 4).

Algorithm 1 ReSPect sampling (per sample): the gate of Sec. 3 with the predictor of Sec. 4.2–4.3; batching stays decoupled.

Table 1: FLUX.1-dev (DrawBench 200, 1024×1024, 50 steps). †SeaCache-reported (RTX PRO 6000); remaining rows ours (B200): compare quality at matched **TFLOPs**. Identical skip gate throughout—only reconstruction differs between the SeaCache and ReSPect rows.

Method	PSNR↑	SSIM↑	LPIPS↓	TFLOPs	Lat(s)
Original (50)	–	–	–	2976	20.9†
TeaCache (δ=0.3)†	20.76	0.810	0.211	1547	11.4
TaylorSeer (S=3)†	22.78	0.828	0.163	1191	9.8
SeaCache (δ=0.3)†	26.29	0.893	0.106	1098	9.4
ReSPect (δ=0.3)	**27.97**	**0.918**	**0.070**	1241	3.33*
TeaCache (δ=0.6)†	17.21	0.714	0.348	892	7.1
TaylorSeer (S=5)†	19.97	0.762	0.236	834	7.5
SeaCache (δ=0.6)†	21.33	0.798	0.226	773	6.4
ReSPect (δ=0.6)	**21.63**	**0.823**	**0.178**	774	2.14*

†SeaCache-reported. *B200. Reference: uncached same-seed output.

Per-sample gating. Threshold crossings are accumulated per sample; batched prompts never share a skip mask. The forecast state per sample is one $(M{+}1) \times F$ coefficient matrix plus K cached residuals: $\mathcal{O}((M{+}1)F + KF)$ memory, and $\mathcal{O}(K(M{+}1)F)$ fitting cost with $M \ll K \ll F$—negligible beside a transformer block (∼0.09 s/image overhead in our runs). The loop is Alg. 1.

5 Experiments

5.1 Settings

Image. FLUX.1-dev, DrawBench (200 prompts), 1024×1024, 50 steps, guidance 3.5, bf16. PSNR/SSIM/LPIPS against the uncached same-seed reference; TFLOPs under the SeaCache convention (2976 at full compute, scaled by computed ratio); $\delta \in \{0.3, 0.6\}$.

Video. HunyuanVideo and Wan2.1-1.3B, VBench prompts, 480p, 65 frames, 50 steps; frame-averaged PSNR/SSIM/LPIPS; $\delta \in \{0.19, 0.35\}$ (Hunyuan), $\{0.2, 0.35\}$ (Wan). Tab. 2 collects SeaCache-reported rows; our Wan rows are complete (Hunyuan running).

Table 2: Video (VBench, 480p, 65f, 50 steps). †SeaCache-reported; *881/822 of 946 rows via MP4-decoded hybrids (∼0.1 dB codec caveat); same-δ comparison, TFLOPs within ∼20%.

Method	PSNR↑	SSIM↑	LPIPS↓	TFLOPs	Lat(s)
		HunyuanVideo			
SeaCache (δ=0.19)†	32.39	0.932	0.047	6747	90.8
SeaCache (δ=0.35)†	26.46	0.857	0.133	4598	58.1
ReSPect (δ=0.19)	TBD	TBD	TBD	TBD	TBD
ReSPect (δ=0.35)	TBD	TBD	TBD	TBD	TBD
		Wan2.1-1.3B			
SeaCache (δ=0.2)†	26.60	0.873	0.075	3942	83.9
SeaCache (δ=0.35)†	21.78	0.740	0.170	2793	56.6
ReSPect (δ=0.2)	28.38*	0.905*	0.066*	4303	12.0*
ReSPect (δ=0.35)	26.70*	0.884*	0.084*	3336	10.0*

5.2 Text-to-image

Tab. 1: at δ=0.3, **+1.68 dB** over the reuse schedule (27.97 vs 26.29) with SSIM 0.918 vs 0.893 and LPIPS 0.070 vs 0.106; at δ=0.6, +0.30 dB (21.63 vs 21.33) at identical 774 TFLOPs. The gain concentrates where the theory predicts: long skips in smooth segments, where (5) is loosest and (7) still tight. Fig. 3 shows both models sit up-left of every published point: higher fidelity at equal or lower compute.

5.3 Text-to-video

On Wan (Tab. 2) the pattern sharpens with skip depth: at δ=0.2, +1.78 dB (28.38 vs 26.60); at δ=0.35, **+4.92 dB** (26.70 vs 21.78) with LPIPS halved (0.084 vs 0.170). Copy-reuse collapses under long skips; the forecast degrades gracefully, as Sec. 4.2 predicts. Fig. 4 checks temporal consistency directly: across four frames of ocean, waterfall, and pool sequences, cached frames track the base trajectory including fine dynamic detail (note the pool splash, bottom right).

5.4 Ablations: testing the analysis

Each ablation targets a prediction of Sec. 4, not a grid. All bars are ΔdB vs copy-reuse on fixed 20-prompt subsets (FLUX δ=0.3, Wan δ=0.35); Fig. 5.

Blend weight w. w=1 discards the low-bias local correction exactly when the global fit is least determined, and collapses: −0.42 dB vs reuse on FLUX, −1.61 dB on Wan. Pure Taylor (w=0) is strong on these short-skip subsets (+1.79/+0.60 dB)—expected, since Taylor is exact locally. The fixed w=0.5 stays above reuse everywhere (+0.65/+0.13 dB): a robust compromise, and the setting behind Tabs. 1–2, rather than a per-subset optimum.

Warm-up threshold. Engaging Chebyshev with $n < M+1$ leaves the fit prior-dominated. Observed: blending from $n \geq 4$ gains +0.65 dB, but the knee sits later—warm-up 12/16 reaches +1.51/+1.77 dB, plateauing at the pure-Taylor limit. Longer Taylor-only spans help on short skips, where locality wins; the spectral term pays off on long skips (cf. Wan δ=0.35, Tab. 2).

Window K. K=100 vs K=32 differs by +0.02 dB: Lemma 1 predicts gains through $\sigma_{\min}$, but per-prompt buffers rarely exceed ∼40 observations, so the window saturates early. K=100 is cheap insurance, not a tuned choice. Residual- rather than state-space forecasting stays a theoretical choice here (Sec. 4.2: the $\|\mathbf{H}\|_F$ constant).

6 Conclusion

Schedules decide where error can be hidden; reconstructors decide how much remains. By moving reconstruction from copying to gated residual spectral prediction—with the blend and warm-up set by analysis rather than search—ReSPect converts the smooth segments a good schedule finds into measured gains at identical compute. Extensions to reward-model and VBench-quality evaluation are underway.

Acknowledgments

We thank the providers of the open-source diffusion and caching implementations this work builds on. Compute was provided by the authors' institution.

References

[1] Chung et al. SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models. CVPR 2026.

[2] Han et al. Adaptive Spectral Feature Forecasting for Diffusion Sampling Acceleration. CVPR 2026.

[3] Liu et al. TeaCache. 2024.

[4] Anonymous. TaylorSeer. 2025.

[5] Selvaraju et al. FORA. 2024.

[6] Huang et al. DiCache. 2025.

[7] Liu et al. AdaCache (video). 2024.

[8] Various. Frequency-aware reuse. 2025.

[9] Zhao et al. PAB. 2024.

[10] Black Forest Labs. FLUX.1. 2024.

[11] Tencent. HunyuanVideo. 2024.

[12] Wan Team. Wan2.1. 2025.

[13] Song et al. Consistency Models. 2023.

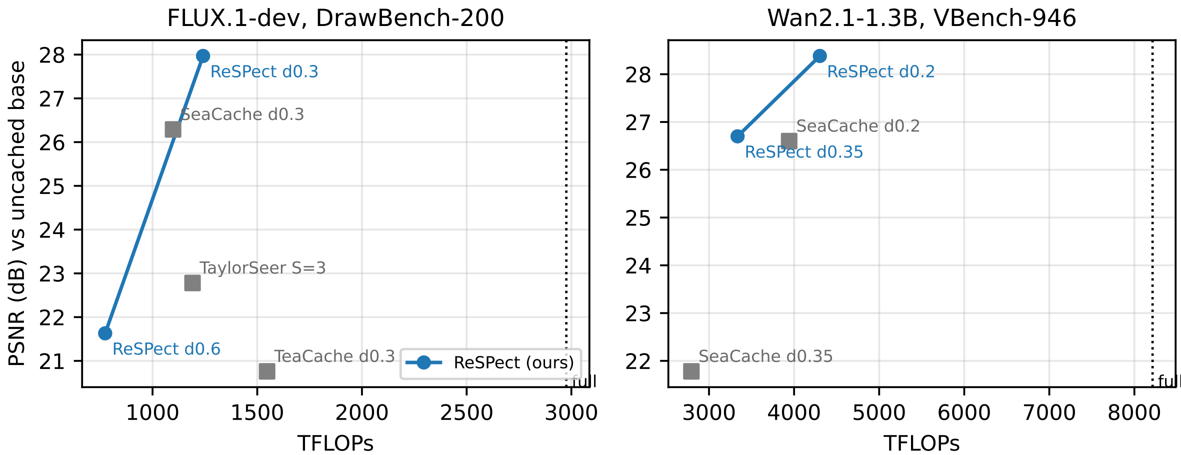

Figure 3: PSNR vs compute. Gray squares: published baselines†; blue: ReSPect. Dotted line: uncached cost. FLUX d0.6 matches SeaCache d0.3's TFLOPs at +0.30 dB; on Wan the d0.35 gap is +4.92 dB.

[14] Yin et al. DMD. 2024.

[15] Song et al. DDIM. 2021.

[16] Lu et al. DPM-Solver. 2022.

[17] Saharia et al. DrawBench. 2022.

[18] Huang et al. VBench. 2024.

[19] Zhang et al. LPIPS. 2018.

[20] Wang et al. SSIM. 2004.

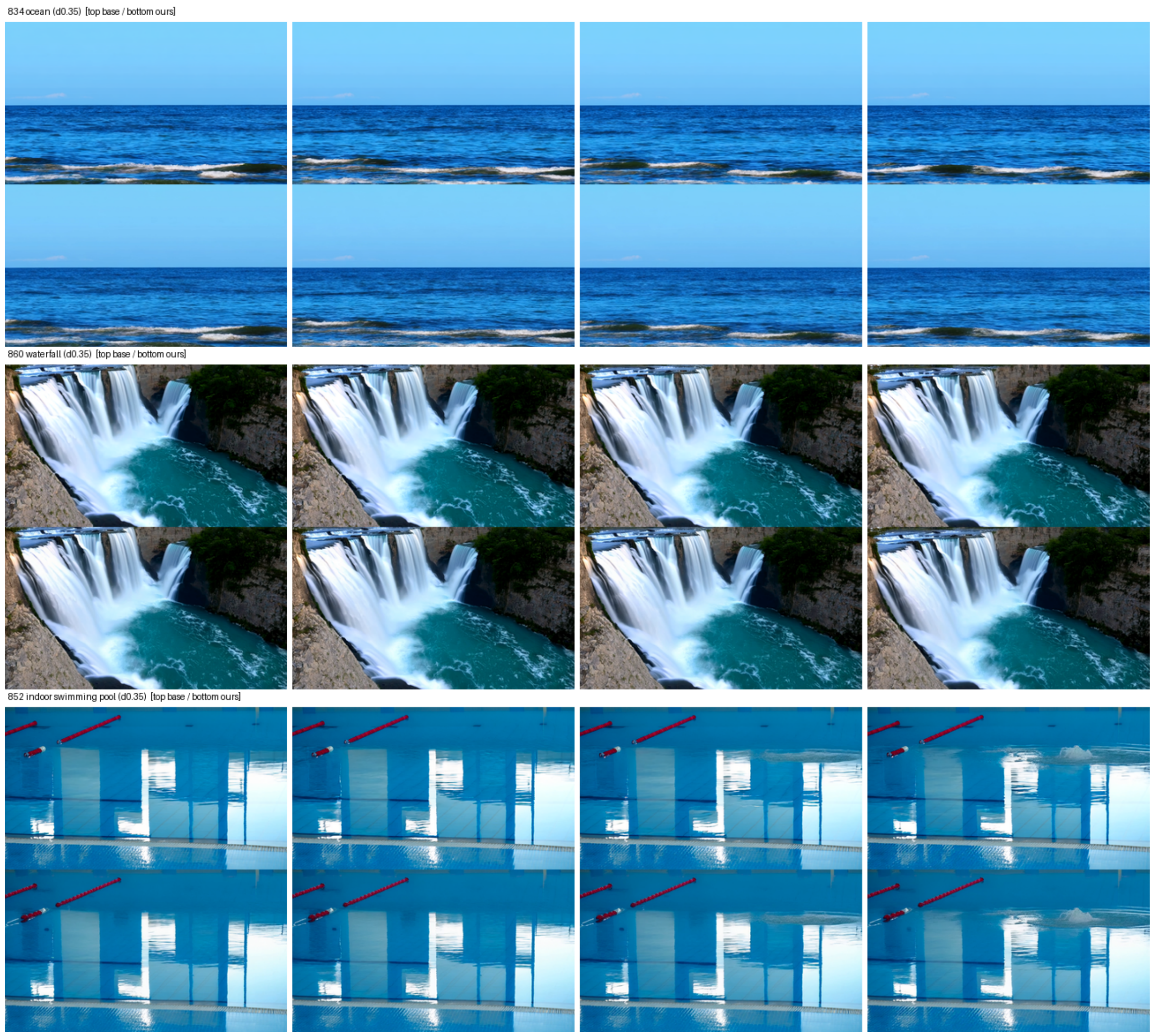

Figure 4: Wan2.1 (δ=0.35) across time: base (top row of each block) vs ReSPect (bottom row). Motion and fine detail persist through $\sim$60% skipped evaluations.

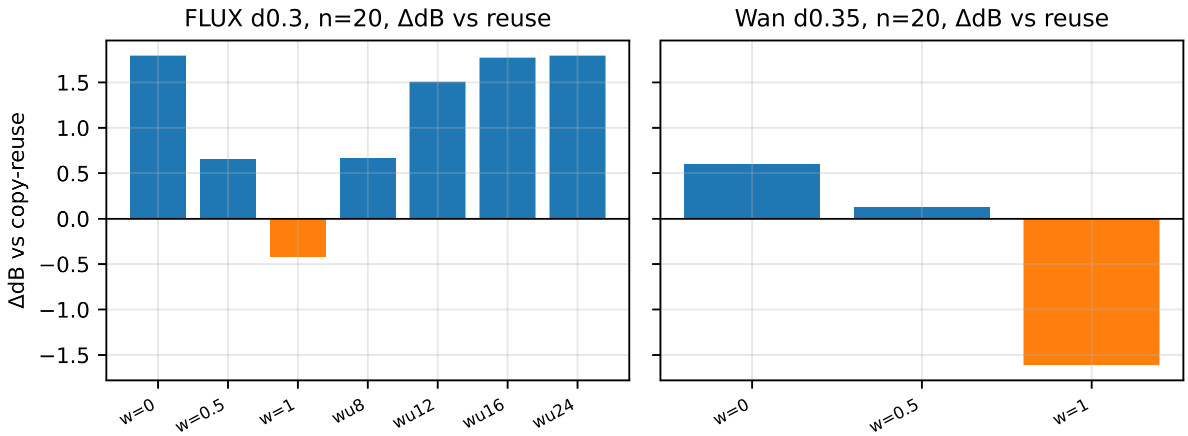

Figure 5: Ablations (ΔdB vs copy-reuse). Left: FLUX δ=0.3. Right: Wan δ=0.35. Pure Chebyshev collapses; pure Taylor wins on short skips; the w=0.5 blend never loses to reuse. Warm-up 12–16 approaches the Taylor limit; window K saturates.

Figure 6: Base (left) vs ReSPect (right): FLUX text-heavy prompt (δ=0.3) and Wan2.1 frames (δ=0.35). Fine detail and composition survive the skipped steps.